FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – XI
PAPER –1
TEST DATE: 28-05-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –06, 19 – 24, 37 – 42):* This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(07 – 10, 25 – 28, 43 – 46):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(11 – 18, 29 – 36, 47 – 54):* This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

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Physics

PART – I

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

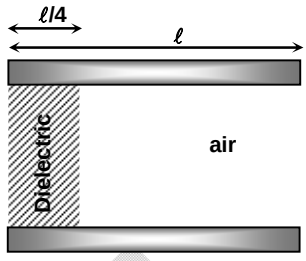
1. A mixture of gases consists of 2g of H_2 gas and 4g of He gas at 102°C. Then, (Take gas constant $R = \frac{25}{3}$ J-mol^{-1}k^{-1})
   (A) The speed of sound wave in the mixture is 975 m/s
   (B) The speed of sound wave in the mixture is 1250 m/s
   (C) If the temperature of the mixture is raised by 7.5°C the percentage change in speed of the sound wave in the mixture is approximately 1%
   (D) If temperature of the mixture is raised by 7.5°C the percentage change in speed of the sound wave in the mixture is approximately 2%

2. An electromagnetic wave is travelling in a non-magnetic medium. The electric field in the medium associated with the electromagnetic wave is given by
   $\vec{E} = (4\hat{i} - 3\hat{j}) \times 10^{-2} \sin(2\pi \times 10^6 z - 4\pi \times 10^{14} t)$ N/C
   Then,
   (A) The average energy density in electric field is $\frac{25\varepsilon_0}{4} \times 10^{-4}$ J/m^3 (where ε_0 is permittivity of the free space)
   (B) The magnetic field $\vec{B}$ associated with the electromagnetic wave is given as
   $\vec{B} = (1.5\hat{i} + 2\hat{j}) \times 10^{-10} \sin(2\pi \times 10^6 z - 4\pi \times 10^{14} t)$T
   (C) The peak value of magnetic field associated with the electromagnetic wave is 2.5×10^{-10} T
   (D) Refractive index of the medium is 2

3. A stream of photons is made to pass through a gaseous sample of identically excited hydrogen like ions. If each photon has energy $\frac{256E_0}{225}$, ten distinct lines are observed in the emission spectrum of the sample. Some of these lines correspond to energy lower than, some higher than and some equal to the energy of the incident photons. Here E_0 is the ionization energy of a hydrogen atom in ground state. Then
   (A) The principle quantum number of the initial excited level of the ions before interaction with the incident photons is n = 2
   (B) The principle quantum number of the highest state of the ions due to interaction of the incident photon is n = 5
   (C) The atomic number of the ion is z = 4
   (D) The minimum energy of the emitted photons after de-excitation of the ion is $\frac{9E_0}{25}$

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4. A parallel plate capacitor of capacitance C_0 is fully charged by a battery of electromotive force V and then disconnected. A dielectric slab covering one-fourth area of each plate is now inserted as shown in the figure. The charge on the capacitor plate gets redistributed such that the region not covered with dielectric and region covered with dielectric share equal amount of charge. Which of the following statement(s) is/are correct?



(A) The dielectric constant of the dielectric is 3

(B) Charge appearing due to polarization on the surface of dielectric is $\frac{1}{3}C_0V$

(C) Force of electrostatic interaction between portion of the plates covered with dielectric equals to that between uncovered portion.

(D) The force of electrostatic interaction between the plates after inserting of the dielectric becomes $\frac{4}{3}$ times of its value before insertion of the dielectric slab.

5. The electric potential inside a spherical volume of radius R varies as $V = ar^2 - br^3$ where r is distance from centre and a and b are constants. Then

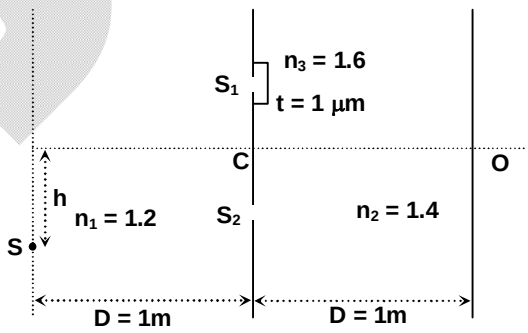
(A) The charge density inside the spherical volume varies as $\rho = 6\varepsilon_0(2br^2 - ar)$

(B) The charge density inside the spherical volume varies as $\rho = 6\varepsilon_0(2br - a)$

(C) At $r = \frac{a}{2b}$, charge density is zero

(D) At $r = \frac{a}{b}$, charge density is maximum

6. In Young's double slit experiment the distance between the slits S_1 and S_2 is $d = 10^{-3}$ m and distance between the plane of slits and the screen is D = 1m. The refractive indices of the media on the left of the slit is $n_1 = 1.2$ and to the right of the plane of slits is $n_2 = 1.4$ as shown in the figure. A point source of light of wavelength 6000Å (in air) is placed at a height h below the line joining the points C and O, where C is mid point of line S_1S_2 and O is the centre of the screen. A thin transparent slab of thickness t = 1 μm and refractive index $n_3 = 1.6$ is place in front of the slit S_1. Then



(A) If h = 0 then intensity of light at the centre of the screen O is 0.50 I_0 where I_0 is maximum fringe intensity on the screen.

(B) If h = 0, the minimum distance of the bright fringe from O is $\frac{1}{7}$mm

(C) If h = d the intensity at point O (centre of the screen) is $\frac{I_0}{4}$

(D) If h = d, the distance of the closest bright fringe from point O is $\frac{2}{7}$mm.

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Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

7. Match the statements in List-I with the values in List-II.

List –I		List –II	
(I)	A stationary source emits sound of frequency 660 Hz. The sound is reflected by a reflector moving away from the source with a velocity 33 m/s. The frequency (in Hz) of the sound reflected by the reflector is (speed of sound in air is 330 m/s)	(P)	600
(II)	A bus is moving towards a hill with speed 16.5 m/s. The driver of the bus sounds a horn of frequency of 627 Hz. The frequency (in Hz) of the echo heard by a passenger in the bus will be (speed of sound in air is 330 m/s)	(Q)	500
(III)	A fighter plane is travelling in a horizontal direction at a certain height from the ground with a speed equal to half the speed of sound in air. The frequency of the sound produced by the plane is 450 Hz. What is the frequency (in Hz) of the sound heard by a stationary observer on the ground when the plane is directly above the observer.	(R)	540
(IV)	A star emits a electromagnetic wave of frequency $500\sqrt{3}$ GHz. It moves away from a observer with a relative speed equal to half of the speed of light in vacuum. The frequency of the wave (in GHz) observed by the observer is	(S)	620
		(T)	693

Which one of the following options is correct ?

(A) $I \rightarrow R,\ II \rightarrow T,\ III \rightarrow S,\ IV \rightarrow P$

(B) $I \rightarrow R\ \ II \rightarrow Q,\ III \rightarrow P\ \ IV \rightarrow S$

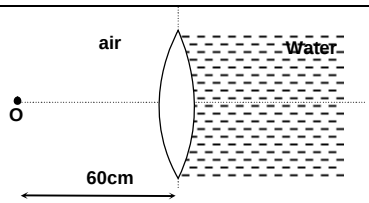
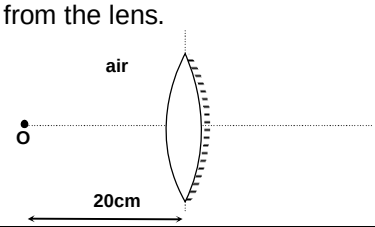
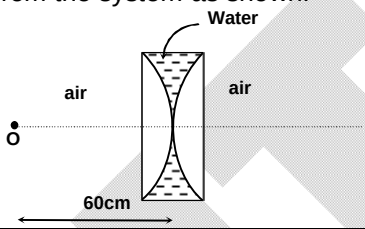
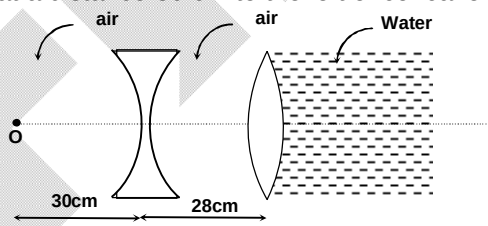
(C) $I \rightarrow P\ \ II \rightarrow T,\ III \rightarrow S\ \ IV \rightarrow Q$

(D) $I \rightarrow R,\ II \rightarrow T,\ III \rightarrow P,\ IV \rightarrow Q$

8. List-I contains the situation of one or more than one thin lenses (refractive index of the material of each lens is 1.5). The refractive indexes of surrounding media are also given in each case. In List-II contains the information of magnitude of final image distance (in cm) in each case. Refractive index of the material of lens in each case is 1.5.

List –I		List –II	
(I)	A thin equiconvex lens of focal length 20 cm (in air) is placed so that medium to the left of it is air (μ_{air} =1) and medium to the right of it is water $\left(\mu_{water} = \frac{4}{3}\right)$. A point object is placed in air on the principal axis of the lens at a distance 60 cm from it.	(P)	240

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(II)	One of the curved surfaces of a thin equiconvex lens of focal length 60 cm (in air), is silvered. A point object is placed to the left of the lens at a distance 20 cm from the lens. 	(Q)	160
(III)	Two thin plano-convex lens each of focal length 30 cm (in air) is placed in contact so that their curved surfaces are in contact. Water $\left(\mu_{water} = \frac{4}{3}\right)$ is filled in the space between the lenses. The media to the left and right side of the system of the lenses is air (μ_{air} = 1). A point object is placed in air at a distance 60 cm from the system as shown. 	(R)	180
(IV)	A concave lens of focal length f_1 = – 20 cm and an equiconvex thin lens of focal length 40 cm are placed at a distance 28 cm so that their principle axes are coinciding. The surroundings media are shown in the diagram. A point object is placed on the principle axis at a distance 30 cm to the left of concave lens 	(S)	80
		(T)	60

Which one of the following options is correct ?

(A) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ R, IV $\rightarrow$ Q

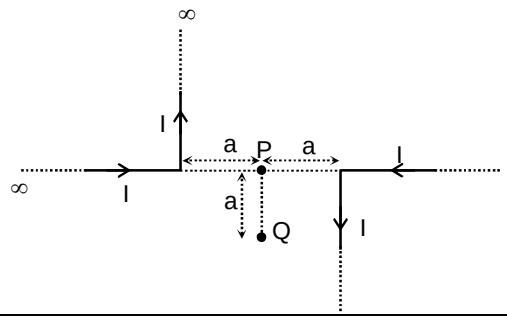
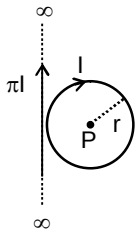
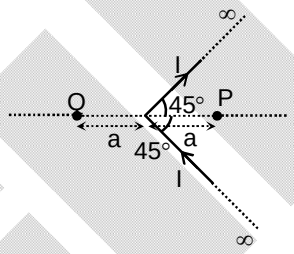
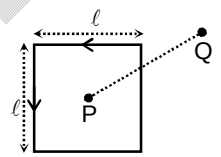
(B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

(C) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ Q

(D) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R

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9. In List-I certain current carrying wires/loops are given and in List-II ratio of magnitudes of magnetic field at two points P and Q are given. Match the List-I with List-II.

List –I		List –II	
(I)		(P)	$\frac{4}{\sqrt{5+2\sqrt{2}}}$
(II)	A circular current carrying loop of radius r and carrying current I and another infinitely long straight current carrying wire carrying current πI are placed very close to each other end that they lie in same plane. The point P is the centre of the circular loop and point Q lies on the normal to the plane of circular loop and passing through the centre of loop at a distance r from it 	(Q)	$\sqrt{6}$
(III)	Two infinitely long straight current carrying wires are placed so that their one of the ends coincide. Two points P and Q lie on the angle bisector of the two wire at same distance from their common end O. 	(R)	1
(IV)	There is a square shaped current carrying loop of side length ℓ and carrying current I. Point P is the centre of the loop and Q lies on the normal to the plane of the loop passing through its centre and a distance ℓ/2 from centre of loop. 	(S)	$\frac{4\sqrt{2}}{\sqrt{5+2\sqrt{2}}}$
		(T)	$3+2\sqrt{2}$

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Which one of the following options is correct ?

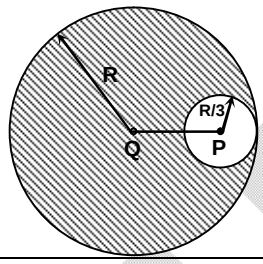
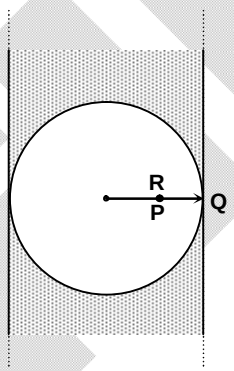
(A) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ R

(B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

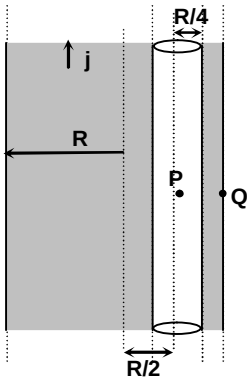
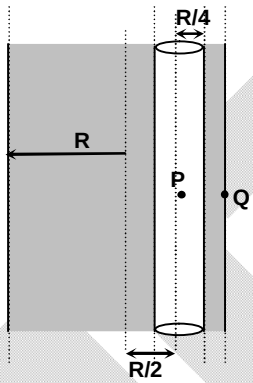
(C) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ T, IV $\rightarrow$ Q

(D) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R

10. In List-I certain charge/current distributions are given. In List-II ratio of magnitude of electric/magnetic fields at two points P and Q (shown in the diagram) are given.

List –I		List –II	
(I)	There is a charged sphere of radius R having spherical cavity of radius R/3. Charge is uniformly distributed inside the sphere except the cavity which is charge free. Let E_P and E_Q be the magnitudes of electric field at P and Q as shown. The ratio $\frac{E_P}{E_Q}$ equals 	(P)	4
(II)	There is an infinitely long cylinder of radius R having a spherical cavity of same radius. Charge is uniformly distributed inside the cylinder except the cavity which is charge free. P is the centre of cavity and Q is point on the surface of cylinder as shown. The magnitude of electric field at point P and Q are E_P and E_Q respectively then $\frac{E_P}{E_Q}$ equals 	(Q)	$\frac{4}{7}$
(III)	There is an infinitely long straight wire with circular cross section of radius R. There is a cylindrical cavity of radius R/4 inside the cylinder. Current is flowing inside the cylinder parallel to its axis with uniform current density j except the cavity. The magnitude of magnetic fields at points P and Q as shown in the figure are B_P and B_Q respectively. The	(R)	3

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	ratio $\frac{B_P}{B_Q}$ equals 		
(IV)	There is an infinitely long cylinder of radius R and cylindrical cavity of radius $\frac{R}{4}$. Charge is uniformly distributed inside the cylinder except the cavity. There is a point P on the axis of the cavity and point Q on the surface of the cylinder as shown. The ratio of magnitudes of electric fields at point P and Q, $\frac{E_P}{E_Q}$ equals 	(S)	8
		(T)	$\frac{1}{2}$

Which one of the following options is correct ?

(A) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ R

(B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

(C) I $\rightarrow$ S, II $\rightarrow$ T III $\rightarrow$ Q, IV $\rightarrow$ Q

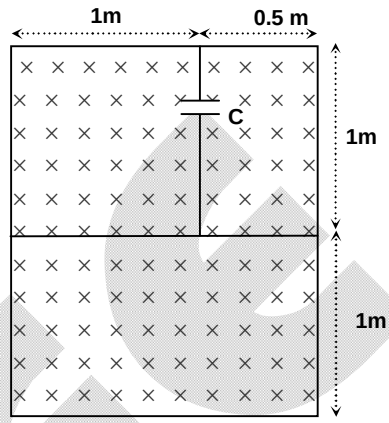
(D) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R

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Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

11. Consider a circuit made of uniform wire. A time varying magnetic field $B = -2t\,\hat{k}$ T is switched on in the entire area of the circuit. The charge (in μC) on the capacitor C = 7.84 μF in the steady state is (The resistance per unit length of the wire is 2 Ω/m)



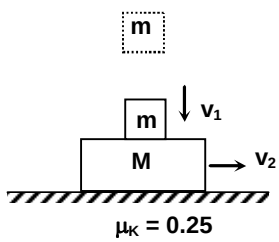
12. Standing wave is set up in an air column (which is closed at one end and open at other end) of length L = 0.99 m. If the air column is vibrating in first overtone find the minimum distance (in m) from the closed end of the air column where the pressure amplitude is $\frac{\sqrt{3}}{2}$ times the maximum pressure amplitude in the column. (neglect end correction)

13. A proton strikes a stationary lithium target and activate a reaction
$^{7}_{3}\text{Li} + ^{1}_{1}\text{p} \rightarrow ^{7}_{4}\text{Be} + ^{1}_{0}\text{n}$
At what value of proton's kinetic energy (in MeV) can the resulting neutron be stationary?
Given mass of $^{7}_{3}\text{Li}$ atom = 7.01601 u, mass of $^{1}_{1}\text{H}$ atom = 1.00783 u, mass of $^{7}_{4}\text{Be}$ atom = 7.01693 u, mass of neutron = 1.00867 u and 1 u $\equiv$ 931.5 MeV

14. A photoelectric plate is initially exposed to an emission spectrum of hydrogen gas excited to first excited state. Later when the same photoelectric plate is exposed to an emission spectrum of some unknown hydrogen like gas, excited to the first excited state, it is found that de-broglie wavelength of the fastest photoelectrons now ejected has decreased to $\sqrt{6.1}$ times. For this new gas difference of energies of first lyman line and series limit of Balmer series is found to be two times the ionization potential of the hydrogen atom. Find the work function (in J) of the photoelectric plate.

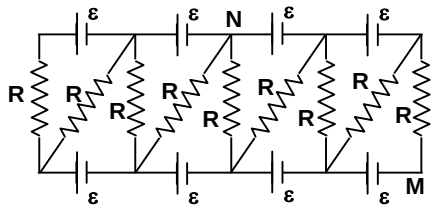
15. A cart moves with uniform velocity on a muddy road. The radius of the wheel is R = 0.6 m. The wheel of the cart does not slip on the road. A small bit of mud detaches from the rim at a height $h = \frac{3R}{2}$ from the ground. Find the speed of cart (in m/s) if the bit of mud falls back on the wheel at the same height (Take g = 10 m/s^2, $\sqrt{3} = 1.73$)

16. A block of mass M = 5kg is moving on a horizontal surface. An object of mass m = 1 kg is dropped onto the block hitting it with a velocity of v_1 = 10 m/s. The speed of the block at the same time instant is v_2 = 2 m/s. The object sticks to the block. The collision time is negligible. What will be the speed of the block after the collision if the coefficient of the kinetic friction between the block and horizontal plane is μ_K = 0.25

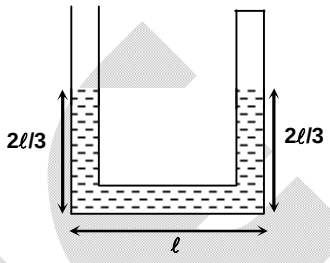


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17. In the circuit shown in the figure each of cell has emf ε = 2.25 V and all the cells are ideal. The value of each resistance R = 2Ω. The magnitude of potential difference (in V) between the points M and N is



18. A thin U-tube sealed at one end consists of three bends of length ℓ = 25 cm each, forming right angles. Mercury is filled in the tube so that the vertical parts of the tube are filled with mercury to height $\frac{2\ell}{3}$. All of mercury can be displaced out of tube by heating slowly the gas in the sealed end of the tube which is separated from the mercury from the atmospheric air by mercury. Find the work done (in J) by the gas. Given atmospherical pressure $p_0 = 10^5$ Pa, density of mercury = 13.6×10^3 kg/m^3 and cross sectional area of the tube S = 0.5 cm^2.



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Chemistry

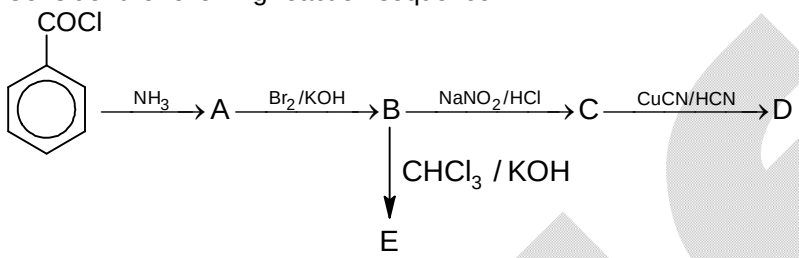
PART – II

Section – A (Maximum Marks: 24)

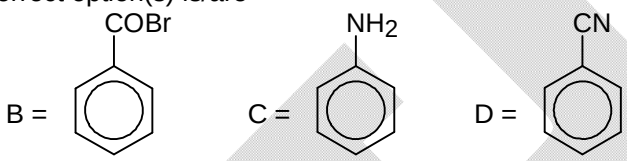
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

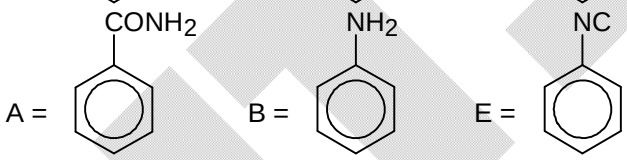
19. The incorrect statement (s) among the following statements is/are
(A) Lactose on hydrolysis produces D-Glucose and D-Mannose
(B) Sucrose undergoes mutarotation
(C) Maltose is a reducing sugar
(D) D-Glucose and D-Mannose are C-4 epimers

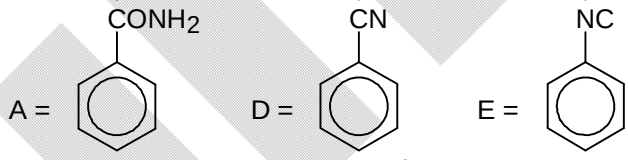
20. Consider the following reaction sequence

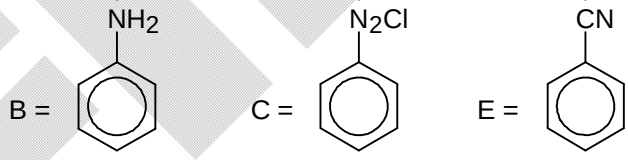


The correct option(s) is/are

(A) 

(B) 

(C) 

(D) 

21. The correct option (s) related to chromyl chloride test is/are
(A) Chromyl Chloride test is carried out for the identification of Cl^-
(B) When Chromyl Chloride vapour is passed into NaOH solution white precipitate is formed
(C) NaCl when heated with $K_2Cr_2O_7$ and conc. H_2SO_4 yellow vapour is evolved
(D) To the solution of Na_2CrO_4 when $Pb(CH_3COO)_2$ is added yellow precipitate is formed

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22. The correct statement (s) related to concentration of ores by Froth Floatation process is/are
(A) Galena is concentrated by Froth Floatation method.
(B) NaCN added to Galena containing ZnS during Froth Floatation acts as a depressant for ZnS.
(C) In Froth Floatation method sodium Cresol is added as a frother.
(D) In Froth Floatation method sodium ethyl xanthate is added as collector.

23. The incorrect statement (s) related to the compounds of phosphorous is/are
(A) Phosphine when passed through $CuSO_4$ solution red precipitate is formed.
(B) PCl_5 in the solid sate exist as $\left[PCl_4^+\right]\left[PCl_6^-\right]$
(C) In the solid state PBr_5 exists as $\left[PBr^+\right]Br^-$
(D) PCl_3 on hydrolysis produces H_3PO_4

24. The correct statement(s) related to colloidal solution is/are
(A) Gold sol is a positive sol.
(B) Ferric hydroxide sol is prepared by peptization.
(C) In the coagulation of a positive sol, the flocculation power of Cl^- is greater than that of PO_4^{3-}
(D) Higher is the gold number higher will be the protecting power of the protective colloid.

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. List – I contains complexes and List – II contains type of hybridisation of central atom and properties of complexes.

List – I		List – II	
(I)	$[FeF_6]^{3-}$	(P)	dsp^2
(II)	$[NiCl_4]^{2-}$	(Q)	sp^3d^2
(III)	$[Ni(CN)_4]^{2-}$	(R)	sp^3
(IV)	$[CoF_6]^{3-}$	(S)	Diamagnetic
		(T)	Paramagnetic

(A) I→Q, S; II→R, S; III→R, S; IV→Q, T
(B) I→Q, T; II→R, T; III→P, S; IV→Q, T
(C) I→Q, T; II→R, T; III→R, S; IV→Q, S
(D) I→Q, S; II→R, T; III→R, S; IV→Q, T

26. List – I contains reaction and List – II contains product of reaction

List – I		List – II	
(I)	$P_4O_{10} + HNO_3 \longrightarrow$	(P)	NO_2
(II)	$Pb(NO_3)_2 \xrightarrow{\Delta}$	(Q)	NO
(III)	$NaNO_2 + FeSO_4 + H_2SO_4 \longrightarrow$	(R)	N_2O_5
(IV)	$NH_4NO_3 \xrightarrow{\Delta}$	(S)	N_2O_3
		(T)	N_2O

(A) I→R; II→Q; III→T; IV→S
(B) I→S; II→P; III→R; IV→Q
(C) I→R; II→S; III→Q; IV→T
(D) I→R; II→P; III→Q; IV→T

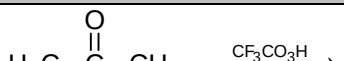
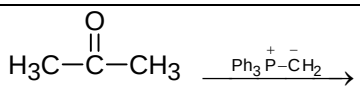
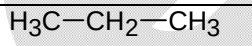
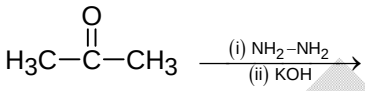
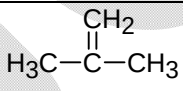
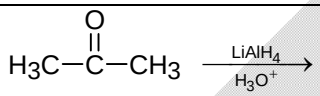
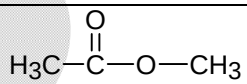
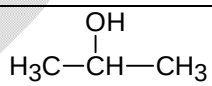
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27. List – I contains reactions and List – II contains colour of the precipitate formed in the reaction gives in List – I:

List – I		List – II	
(I)	$Cu(NO_3)_2 + K_4[Fe(CN)_6] \longrightarrow$	(P)	Black ppt.
(II)	$Pb(NO_3)_2 + K_2CrO_4 \longrightarrow$	(Q)	Yellow ppt.
(III)	$BiCl_3 + H_2O \longrightarrow$	(R)	Green ppt.
(IV)	$Pb(CH_3COO)_2 + H_2S \longrightarrow$	(S)	White ppt.
		(T)	Chocolate brown ppt.

(A) I→T; II→Q; III→S; IV→P
(B) I→T; II→Q; III→R; IV→P
(C) I→Q; II→P; III→T; IV→S
(D) I→P; II→R; III→S; IV→T

28. List – I contains reactions and List – II contains products of the reaction in List – I:

List – I		List – II	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	

(A) I→T; II→R; III→P; IV→Q
(B) I→S; II→R; III→Q; IV→T
(C) I→S; II→Q; III→P; IV→T
(D) I→T; II→R; III→Q; IV→R

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

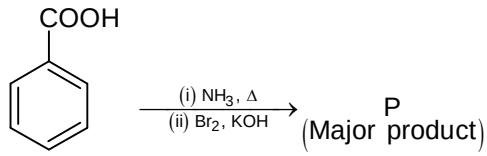
29. 'W' gram of urea (MW = 60) when dissolved in 45 gram water, the vapour pressure of water decreases by 20%. The value of 'W' is _____

30. Standard molar heat of combustion of ethane is 1564.5 kJ. Standard heat of formation of CO_2 and H_2O are -395 kJ and -286 kJ respectively. The standard heat of formation of ethane is $-x$ kJ. The value of x is

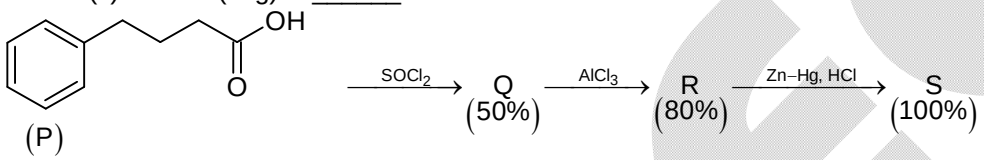
31. Vapour density of $N_2O_4(g)$ at certain temperature is 32. The percentage dissociation of $N_2O_4(g)$ at this temperature is _____%

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32. 40 ml of $\frac{M}{10}$ M CH_3COOH solution is mixed with 32 ml of $\frac{M}{10}$ NaOH solution. pK_a of CH_3COOH is 4.76. The pH of resulting solution is _____
(log 2 = 0.3)

33. 

The weight percentage of N in the compound P is
(Given atomic mass of H = 1, C = 12, N = 14)

34. In the reaction sequence given below is carried out with one mole of reactant (P), the amount of product (s) formed (in g) is _____


The yield of Q, R and S are given in parentheses?

35. 5 g of a sample of impure MnO_2 was heated with excess concentrated HCl and the evolved Cl_2 was passed through excess KI solution. The liberated I_2 required 30 ml of 0.1 M $Na_2S_2O_3$. The percentage of MnO_2 in the sample of impure MnO_2 is ____

36. 4.42 g of Barium azide on complete decomposition gives a colourless gas P. The mass (in gram) of magnesium required to completely react on heating with the evolved gas 'P' is_____
(Given atomic mass of Ba = 137, N = 14 and Mg = 24)

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Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

37. Let $f : [0, \infty) \to [0, \infty)$, $f(x) = \lambda(e^{-x} - e^{x})$, $\lambda \geq 0$, then $\lim\limits_{x\to 3}\dfrac{f(x)-f(3)}{x-3}$ is equal to

(A) λ (B) 2λ

(C) $\lambda\left(e^{-3}+e^{3}\right)$ (D) $-\lambda\left(e^{-3}+e^{3}\right)$

38. Let a rectangular hyperbola passes through the points A(1, 0), B(3, 4) and C(5, 3) equation of tangent at B on the given hyperbola can be

(A) x + 2y = 11 (B) 4x + 3y = 24

(C) x + 3y = 15 (D) none of these

39. If roots of $z^2 + \alpha z + \beta = 0$ lie on $|z| = 2$, then which of the following may be correct ($\alpha, \beta \in C$)

(A) $|\alpha| = 3$ (B) $|\beta| = 4$

(C) $|\alpha| = 2$ (D) $|\beta| = 3$

40. If $\int\limits_0^1 {}^{207}C_7 x^{200}(1-x)^7\,dx = \dfrac{1}{16k}$, then

(A) k is odd integer (B) k is prime

(C) k is divisible by 3 (D) k is even integer

41. Let A = {1, 2, 3, ..., n} and B = {2023, 2024, 2025}. If k_n denotes the total number of functions from A to B such that sum of entries in range of the function is odd, then

(A) $\dfrac{(k_5)}{2}$ is prime (B) $k_5 > 100$

(C) sum of digits of $k_7 = 11$ (D) $k_6 + k_7 = 380$

42. Let $p = x^2 - 3x + 2$, $0 \leq x \leq 2$ and $r_0 \leq p \leq d_0$ for the given values of x. If r is the maximum value of r_0 and d is the minimum value of d_0 and $S = \lim\limits_{n\to\infty}\sum\limits_{k=0}^{n}(3+kd)r^k$, then

(A) $4r + d - 1 = 0$ (B) $S = \dfrac{52}{25}$

(C) $d = -2r$ (D) $S = \dfrac{26}{25}$

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Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. Let S be the set of matrices as $S=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ such that abcd denotes all possible 4 digit decimal number. For some $A=\begin{bmatrix} p & q \\ r & s \end{bmatrix} \in S$, D(A) = ps – qr, T(A) = p + s, T′(A) = q + r

List-I		List-II	
(I)	Product of number of singular matrices in S which have no zero entry and number of singular matrices in S which have atleast one zero entry	(P)	20250
(II)	$\sum_{A\in S} D(A)$	(Q)	85500
(III)	$\sum_{A\in S} T(A)$	(R)	35739
(IV)	$\sum_{A\in S} T'(A)$	(S)	81000
		(T)	none of these

The correct option is:
(A) (I) → (P); (II) → (Q); (III) → (R); (IV) → (S)
(B) (I) → (R); (II) → (P); (III) → (S); (IV) → (Q)
(C) (I) → (R); (II) → (P); (III) → (Q); (IV) → (S)
(D) (I) → (P); (II) → (Q); (III) → (S); (IV) → (R)

44. Match the following List-I with List-II

List-I		List-II	
(I)	If the function $y=e^{4x}+2e^{-x}$ is a solution of the differential equation $\frac{\frac{d^3y}{dx^3}-13\frac{dy}{dx}}{y}=k$, then the value of $\frac{k}{3}$ is	(P)	3
(II)	Number of straight lines which satisfy the differential equation $\frac{dy}{dx}+x\left(\frac{dy}{dx}\right)^2-y=0$ is/are	(Q)	4
(III)	If the substitution $y = t^m$ will transform the differential equation $2x^4y\frac{dy}{dx}+y^4=4x^6$ into a homogeneous equation, then the value of 2m is	(R)	2
(IV)	If the solution of the differential equation $x^2\frac{d^2y}{dx^2}+2x\frac{dy}{dx}=12y$ is $y=Ax^m+Bx^{-n}$, then |m – n| can be	(S)	1
		(T)	0

The correct option is:

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(A) (I) → (Q); (II) → (R); (III) → (P); (IV) → (S)
(B) (I) → (Q); (II) → (P); (III) → (R); (IV) → (S)
(C) (I) → (R); (II) → (Q); (III) → (P); (IV) → (S)
(D) (I) → (R); (II) → (Q); (III) → (S); (IV) → (P)

45. A bag contains some white and some black balls, all combinations being equally likely. The total number of balls in the bag is 12. Four balls are drawn at random from the bag without replacement

List-I		List-II	
(I)	Probability that all four balls are black is equal to	(P)	$\frac{14}{33}$
(II)	If the bag contains 10 black and 2 white balls, then the probability that all four balls are black is equal to	(Q)	$\frac{1}{5}$
(III)	If all the four balls are black, then the probability that the bag contains 10 black balls is equal to	(R)	$\frac{70}{429}$
(IV)	Probability that two balls are black and two are white is equal to	(S)	$\frac{13}{165}$
		(T)	none of these

The correct option is:
(A) (I) → (P); (II) → (Q); (III) → (R); (IV) → (S)
(B) (I) → (R); (II) → (R); (III) → (Q); (IV) → (P)
(C) (I) → (Q); (II) → (P); (III) → (R); (IV) → (Q)
(D) (I) → (Q); (II) → (T); (III) → (Q); (IV) → (S)

46. Match the following List-I with List-II

List-I		List-II	
(I)	If $p^2 + 9q^2 + 25r^2 = pqr\left(\frac{15}{p}+\frac{5}{q}+\frac{3}{r}\right)$ and p, q, r are real, then	(P)	p = q = r
(II)	$\log_{10}(p+r), \log_{10}(r-p), \log_{10}(p-2q+r)$ are in A.P., then	(Q)	$p \neq q \neq r$
(III)	Let $f(x) = px^2 + qx + r$ be a polynomial function of second degree. If f(1) = f(–1) and f′(p), f′(q), f′(r) are in A.P., then	(R)	2q = p + r
(IV)	Three positive numbers p, q, r satisfy (i) p + q + r = 25 (ii) 2 < p, q, r < 18 (iii) 2p = q + 2 (iv) $r^2 = 18q$, then	(S)	$q^2 = pr$
		(T)	p, q, r are in H.P.

The correct option is:
(A) (I) → (Q); (II) → (R); (III) → (P); (IV) → (R)
(B) (I) → (T); (II) → (R); (III) → (R); (IV) → (S)
(C) (I) → (R); (II) → (T); (III) → (Q); (IV) → (R)
(D) (I) → (T); (II) → (T); (III) → (R); (IV) → (Q)

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Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

47. Let normal to the parabola $y^2 = 8x$ at P meet the curve again at Q. If PQ and the normal at Q makes angle α and β respectively with the x-axis, then $|\tan\alpha(\tan\alpha + \tan\beta)|$ is equal to

48. If $\frac{\cos\theta_1}{\cos\theta_2}+\frac{\sin\theta_1}{\sin\theta_2}=\frac{\cos\theta_0}{\cos\theta_2}+\frac{\sin\theta_0}{\sin\theta_2}=1$ (where $\theta_1 - \theta_0 \neq (2n+1)\pi$, $n \in I$), then the value of $\left|\frac{\cos\theta_1\cos\theta_0}{\cos^2\theta_2}+\frac{\sin\theta_1\sin\theta_2}{\sin^2\theta_2}\right|$ is

49. Let $S=\sum_{n=1}^{9999}\frac{1}{\left(\sqrt{n}+\sqrt{n+1}\right)\left(\sqrt[4]{n}+\sqrt[4]{n+1}\right)}$, then the value of S is

50. If the equation $3\sqrt{-2(x+3)}-1=|x+3|+k$ has exactly two real roots, then the maximum possible value of $|[k]|$ is (where [.] represents greatest integer function)

51. If the value of $\int_0^1\frac{\sin^{-1}\left(\sqrt{x}\right)}{x^2-x+1}dx=\frac{\pi^2}{3\sqrt{3n}}$ (where $n \in N$), then n is equal to

52. If x satisfies the cubic equation $ax^3 + bx^2 + cx + d = 0$ such that $\cos^{-1}(x) + \cos^{-1}(2x) + \cos^{-1}(3x) = \pi$, then the value of $|(b + c) – (a + d)|$ is equal to (HCF (a, b) = 2, a, b, c, d are integers)

53. If $\lim_{x\to0}\frac{x^3\left(1-x^2\right)^{1/2}+3\left(1-x^3\right)^{1/3}-3\left(1-x^4\right)^{1/4}}{x^4}=L$ then $|12L|$ is

54. If the value of $\sum_{r=1}^{49}\frac{2r^2-48r+1}{(50-r)\,{}^{50}C_r}=k$, then the value of [k] is (where [.] represents greatest integer function)

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